

INSTITUTO FEDERAL DE EDUCAÇÃO CIÊNCIA E TECNOLOGIA
DE MINAS GERAIS - *CAMPUS* SÃO JOÃO EVANGELISTA
BACHARELADO EM SISTEMAS DE INFORMAÇÃO

Valdir de Souza Carvalho Neto

Desafio 02 - Configuração do DNS

Redes de Computadores III

São João Evangelista

2026

LISTA DE ILUSTRAÇÕES

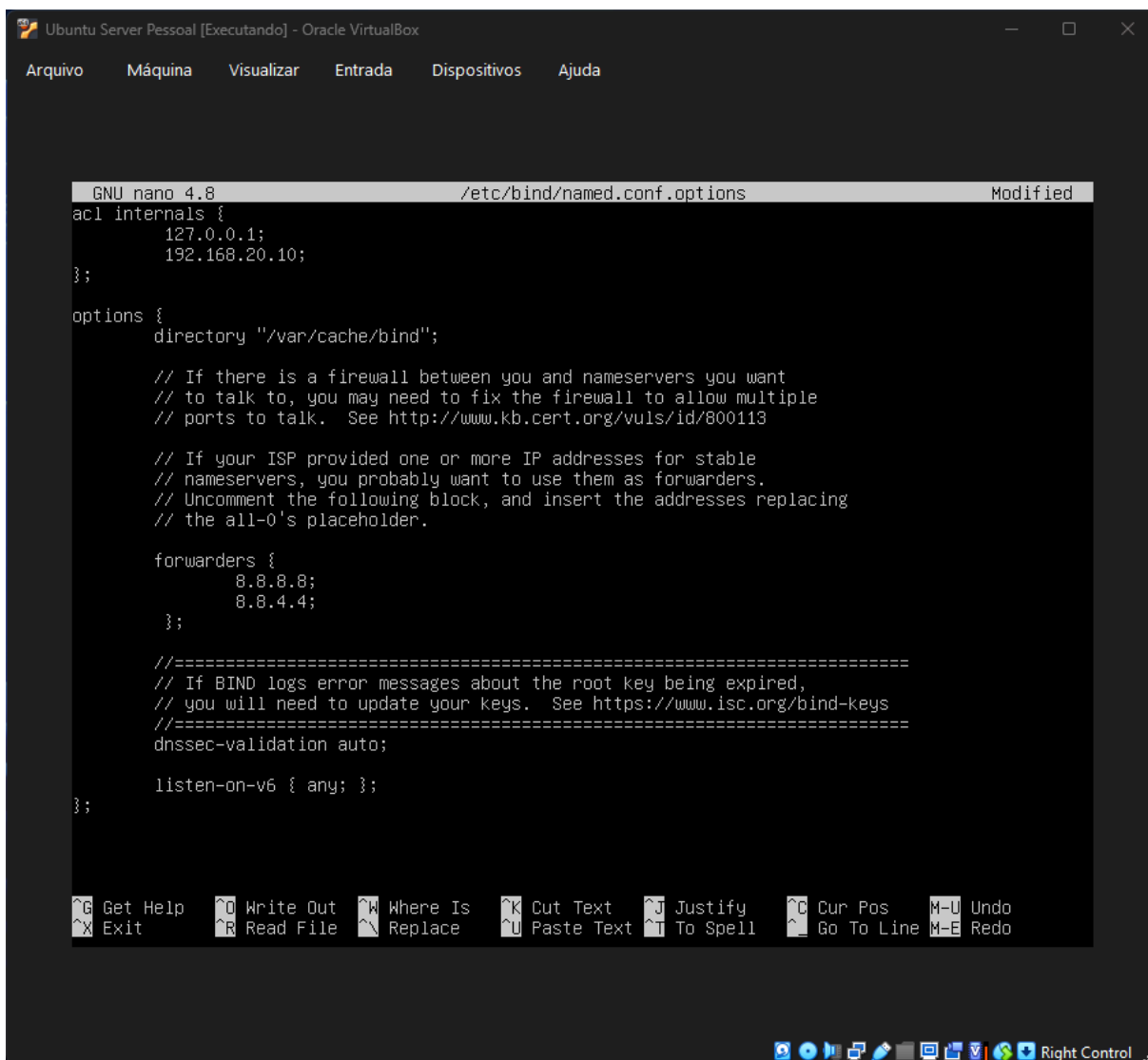
Figura 1 – Conteúdo do arquivo de opções <code>named.conf.options</code>	4
Figura 2 – Declaração das zonas no arquivo <code>named.conf.local</code>	5
Figura 3 – Registros de mapeamento na zona de pesquisa direta.....	6
Figura 4 – Registros de ponteiros (PTR) na zona de pesquisa reversa.....	7
Figura 5 – Checagem de sintaxe dos arquivos estruturais com o <code>named-checkconf</code>	8
Figura 6 – Validação das zonas direta e reversa com o <code>named-checkzone</code>	9
Figura 7 – Verificação do status de execução do serviço <code>bind9</code>	10
Figura 8 – Testes de resolução direta e reversa com o comando <code>nslookup</code> no cliente.....	11

SUMÁRIO

1. SERVIDOR DO DNS.....	4
2. CLIENTE XP.....	11

1. SERVIDOR DO DNS

Figura 1 – Conteúdo do arquivo de opções *named.conf.options*



The screenshot shows a terminal window titled "Ubuntu Server Pessoal [Executando] - Oracle VirtualBox". Inside the terminal, the GNU nano 4.8 editor is open, displaying the file /etc/bind/named.conf.options. The file content is as follows:

```
GNU nano 4.8 /etc/bind/named.conf.options Modified
acl internals {
    127.0.0.1;
    192.168.20.10;
};

options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    forwarders {
        8.8.8.8;
        8.8.4.4;
    };

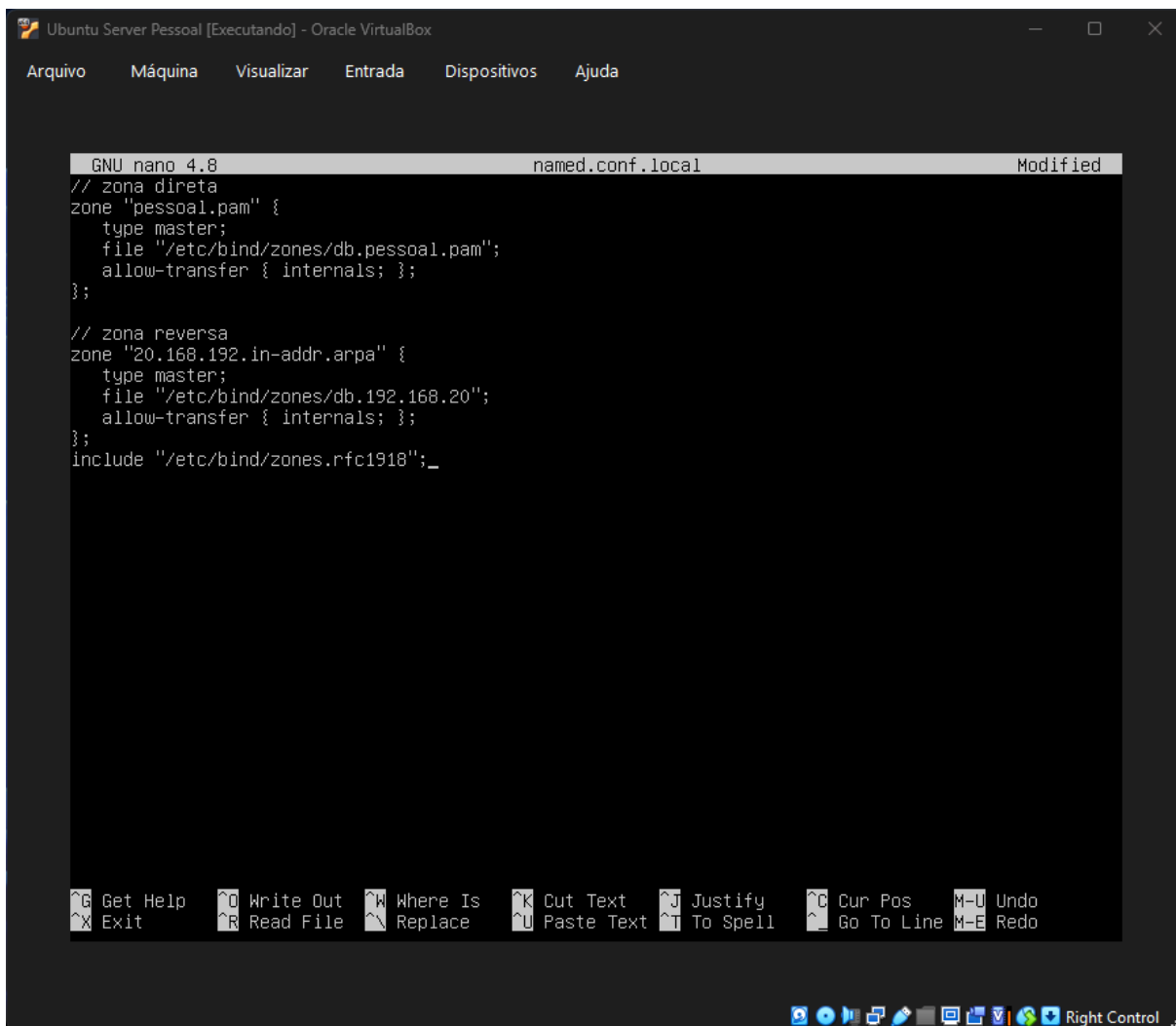
    //=====
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.isc.org/bind-keys
    //=====
    dnssec-validation auto;

    listen-on-v6 { any; };
};
```

The bottom of the terminal shows the nano editor's status bar with various keyboard shortcuts: ^G Get Help, ^O Write Out, ^W Where Is, ^K Cut Text, ^J Justify, ^C Cur Pos, M-U Undo, ^X Exit, ^R Read File, ^_ Replace, ^U Paste Text, ^T To Spell, ^_ Go To Line, M-E Redo. The system's taskbar is visible at the bottom of the window.

Fonte: Elaborado pelo autor (2026).

Figura 2 – Declaração das zonas no arquivo *named.conf.local*



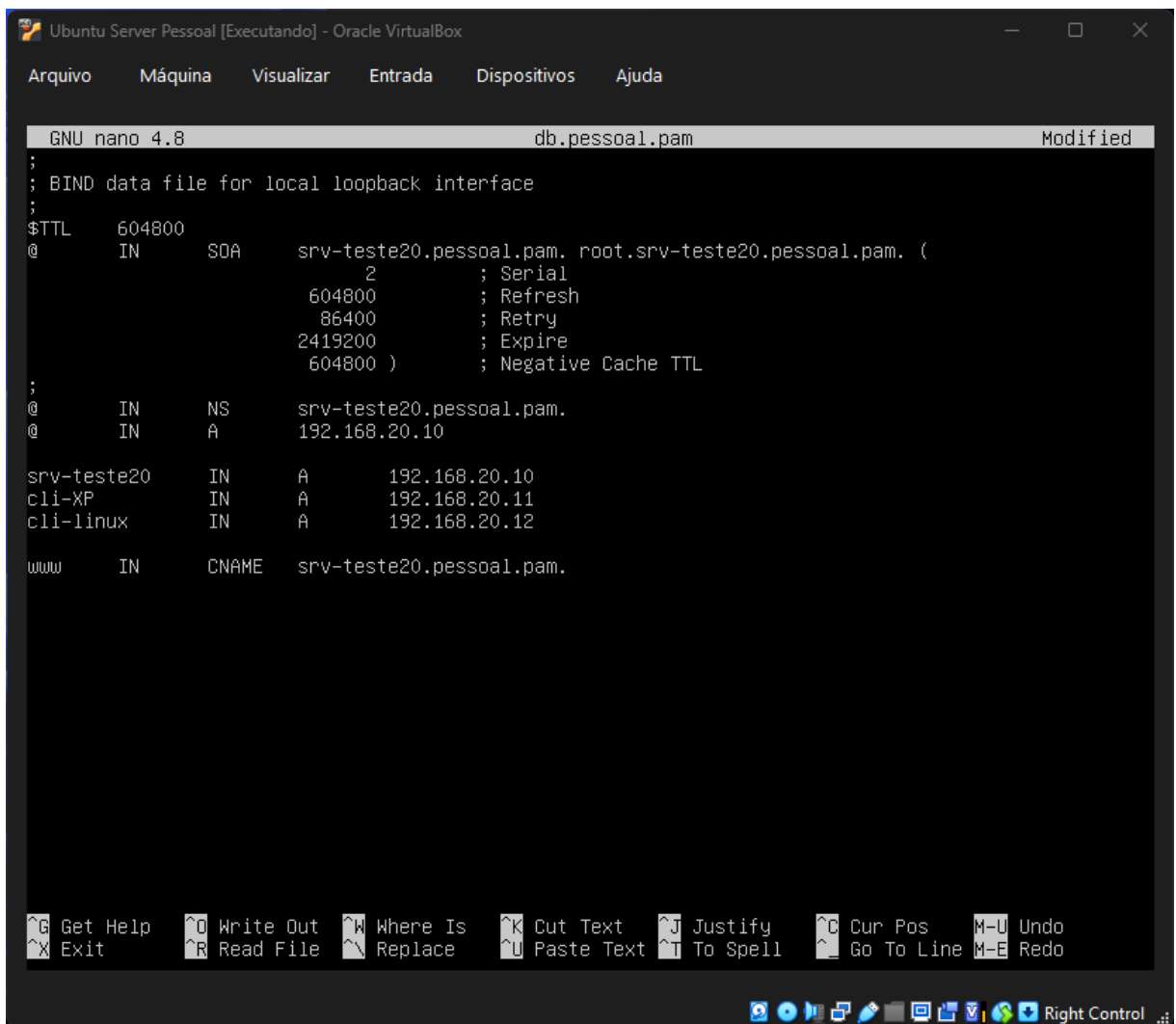
```
GNU nano 4.8                                named.conf.local          Modified
// zona direta
zone "pessoal.pam" {
    type master;
    file "/etc/bind/zones/db.pessoal.pam";
    allow-transfer { internals; };
};

// zona reversa
zone "20.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/zones/db.192.168.20";
    allow-transfer { internals; };
};
include "/etc/bind/zones.rfc1918";_

^G Get Help    ^O Write Out   ^H Where Is    ^K Cut Text    ^J Justify     ^C Cur Pos     M-U Undo
^X Exit        ^R Read File   ^W Replace     ^U Paste Text  ^T To Spell    ^_ Go To Line   M-E Redo
```

Fonte: Elaborado pelo autor (2026).

Figura 3 – Registros de mapeamento na zona de pesquisa direta



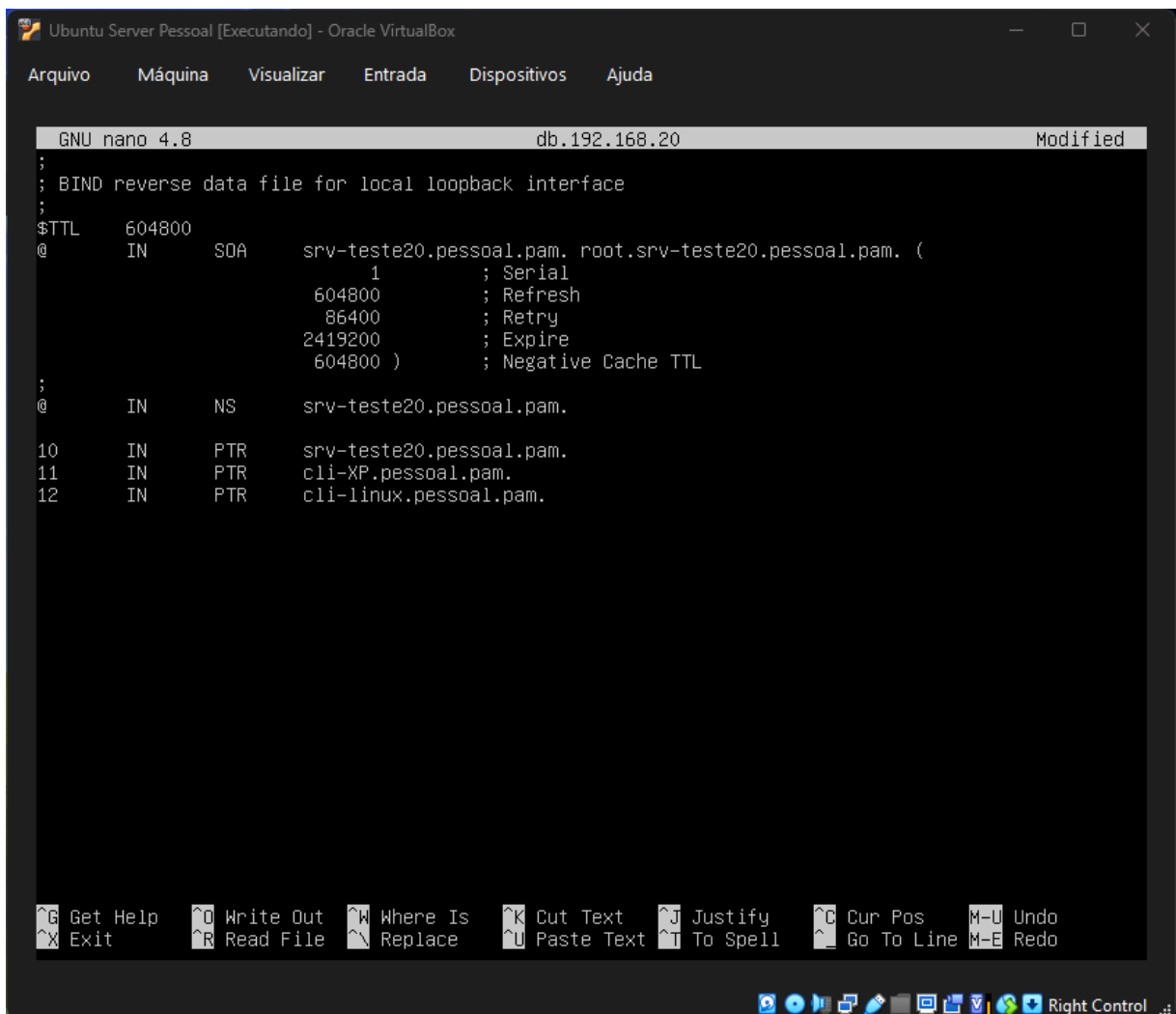
The image shows a terminal window titled "Ubuntu Server Pessoal [Executando] - Oracle VirtualBox". Inside the terminal, the GNU nano 4.8 editor is open, editing the file "db.pessoal.pam". The file contains DNS zone data for "srv-teste20.pessoal.pam". The records include an SOA record, an NS record, and three A records for "cli-xp", "cli-linux", and "www". The bottom of the terminal shows a menu with various editing and navigation commands.

```
GNU nano 4.8 db.pessoal.pam Modified
;
; BIND data file for local loopback interface
;
$TTL 604800
@ IN SOA srv-teste20.pessoal.pam. root.srv-teste20.pessoal.pam. (
    2 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS srv-teste20.pessoal.pam.
@ IN A 192.168.20.10
srv-teste20 IN A 192.168.20.10
cli-xp IN A 192.168.20.11
cli-linux IN A 192.168.20.12
www IN CNAME srv-teste20.pessoal.pam.
```

Menu: ^G Get Help, ^O Write Out, ^W Where Is, ^K Cut Text, ^J Justify, ^C Cur Pos, M-U Undo, ^X Exit, ^R Read File, ^_ Replace, ^U Paste Text, ^T To Spell, ^_ Go To Line, M-E Redo

Fonte: Elaborado pelo autor (2026).

Figura 4 – Registros de ponteiros (PTR) na zona de pesquisa reversa



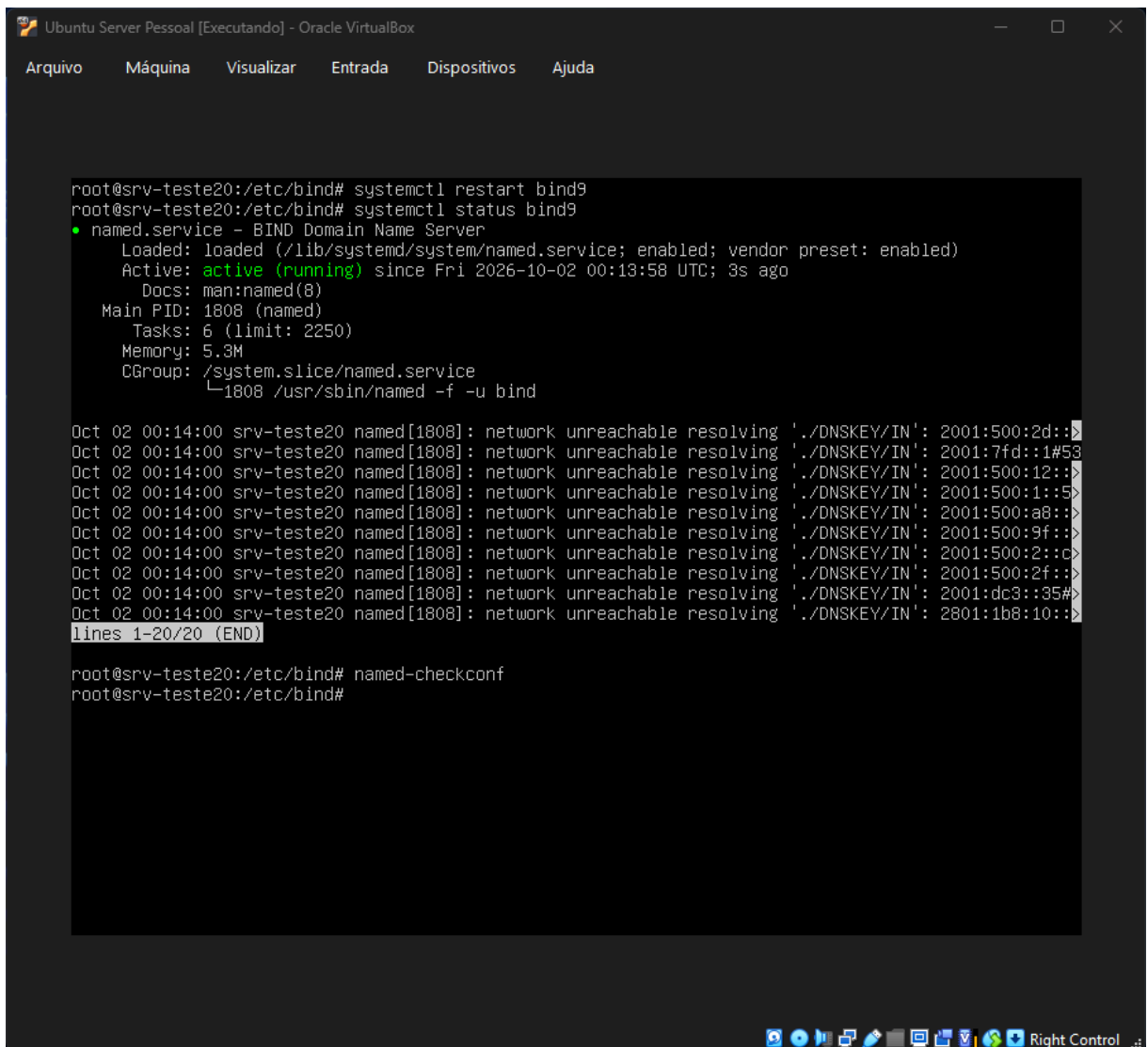
The screenshot shows a terminal window titled "Ubuntu Server Pessoal [Executando] - Oracle VirtualBox". Inside, the nano 4.8 text editor is open, displaying a BIND reverse data file for the local loopback interface. The file content is as follows:

```
GNU nano 4.8 db.192.168.20 Modified
;
; BIND reverse data file for local loopback interface
;
$TTL 604800
@ IN SOA srv-teste20.pessoal.pam. root.srv-teste20.pessoal.pam. (
    1 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS srv-teste20.pessoal.pam.
10 IN PTR srv-teste20.pessoal.pam.
11 IN PTR cli-XP.pessoal.pam.
12 IN PTR cli-linux.pessoal.pam.
```

The bottom of the terminal shows the nano editor's command palette with various shortcuts like ^G Get Help, ^O Write Out, ^W Where Is, ^K Cut Text, ^J Justify, ^C Cur Pos, M-U Undo, ^X Exit, ^R Read File, ^_ Replace, ^U Paste Text, ^T To Spell, ^_ Go To Line, M-E Redo, and a Right Control key indicator.

Fonte: Elaborado pelo autor (2026).

Figura 5 – Checagem de sintaxe dos arquivos estruturais com o *named-checkconf*



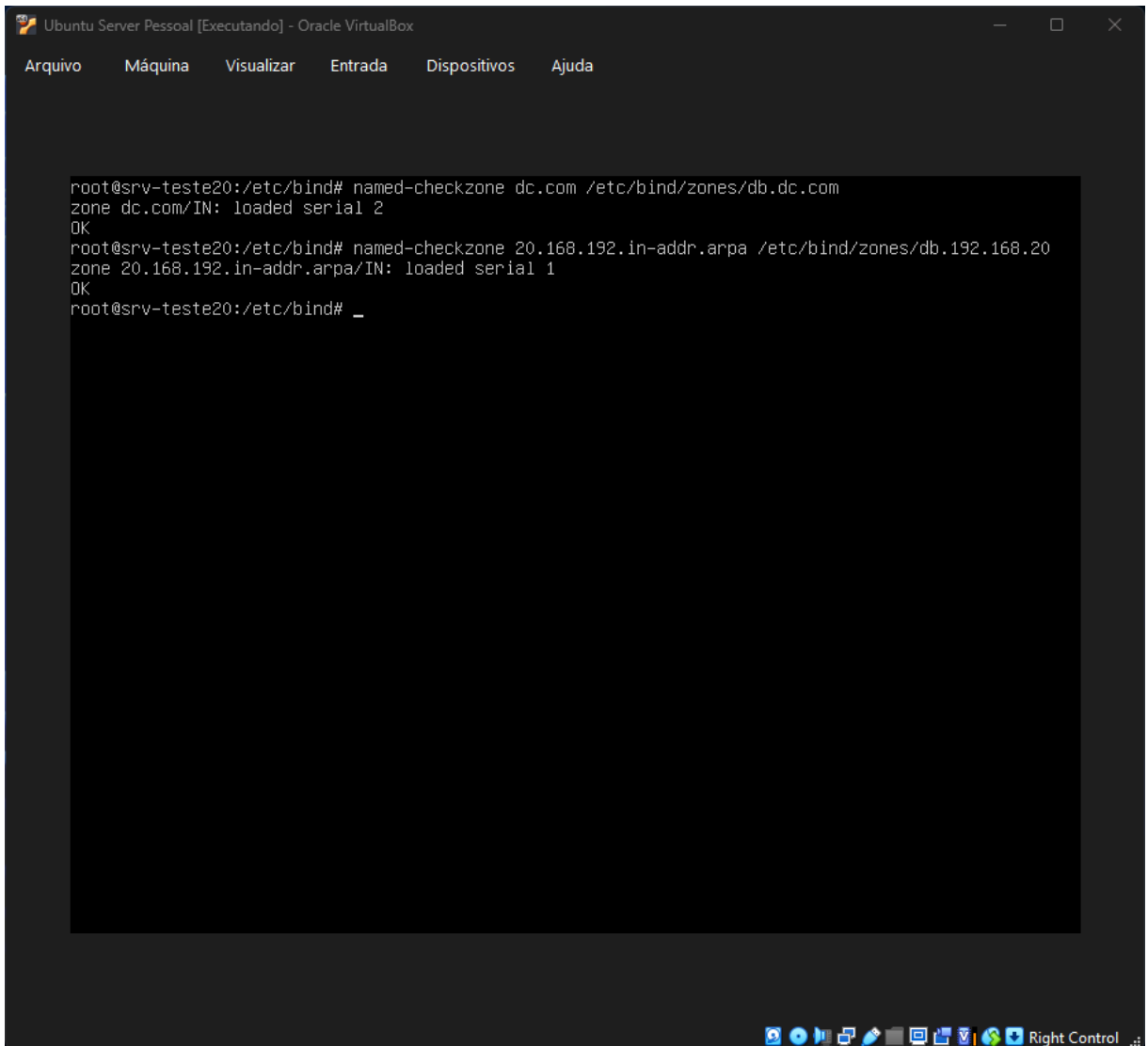
```
root@srv-teste20:/etc/bind# systemctl restart bind9
root@srv-teste20:/etc/bind# systemctl status bind9
• named.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-10-02 00:13:58 UTC; 3s ago
     Docs: man:named(8)
    Main PID: 1808 (named)
      Tasks: 6 (limit: 2250)
     Memory: 5.3M
    CGroup: /system.slice/named.service
            └─1808 /usr/sbin/named -f -u bind

Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2001:500:2d::>
Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2001:7fd::1#53
Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2001:500:12::>
Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2001:500:1::5>
Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2001:500:a8::>
Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2001:500:9f::>
Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2001:500:2::c>
Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2001:500:2f::>
Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2001:dc3::35#>
Oct 02 00:14:00 srv-teste20 named[1808]: network unreachable resolving './DNSKEY/IN': 2801:1b8:10::>
lines 1-20/20 (END)

root@srv-teste20:/etc/bind# named-checkconf
root@srv-teste20:/etc/bind#
```

Fonte: Elaborado pelo autor (2026).

Figura 6 – Validação das zonas direta e reversa com o *named-checkzone*



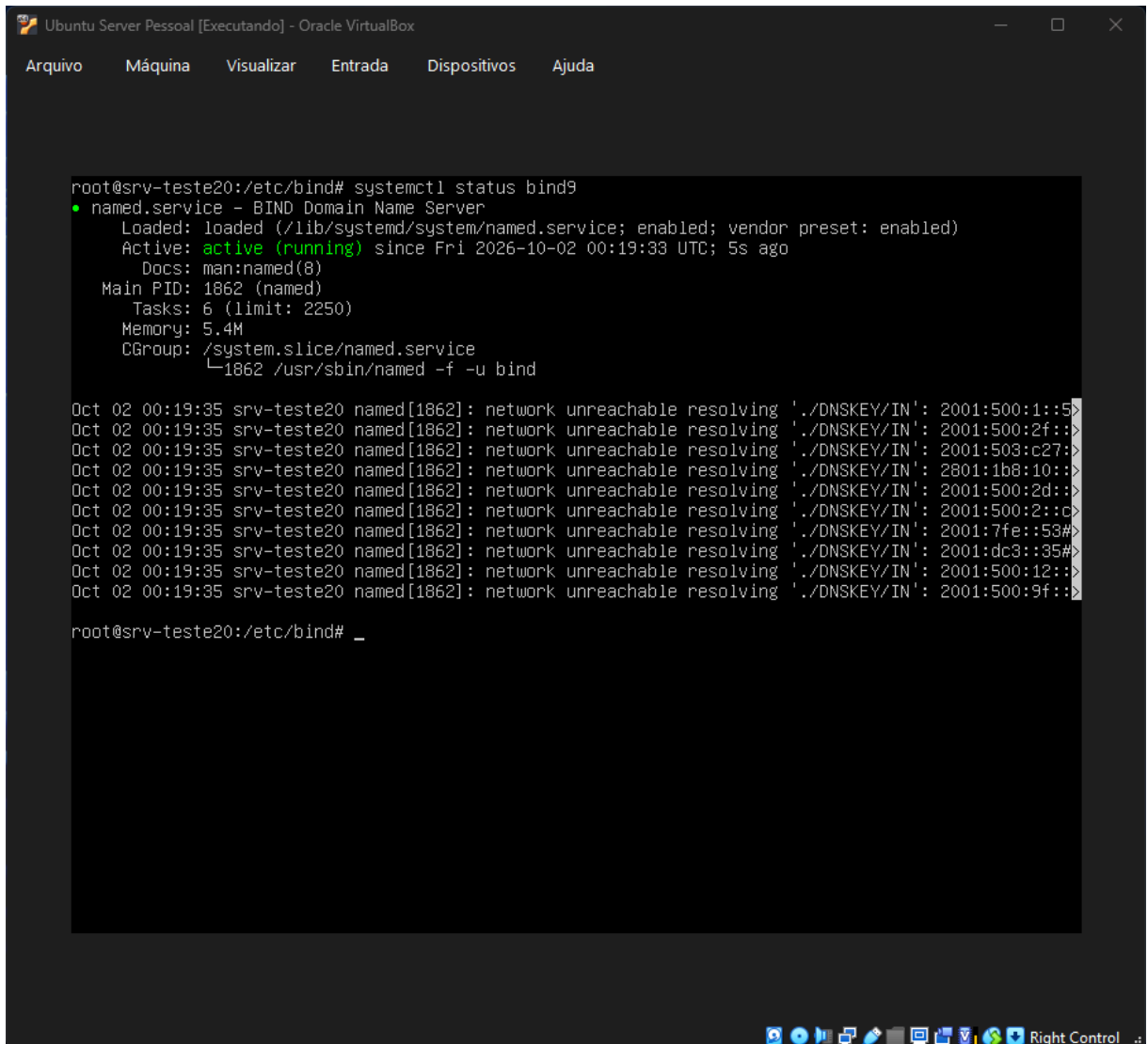
The image shows a terminal window titled "Ubuntu Server Pessoal [Executando] - Oracle VirtualBox". The window has a menu bar with "Arquivo", "Máquina", "Visualizar", "Entrada", "Dispositivos", and "Ajuda". The terminal output shows the following commands and results:

```
root@srv-teste20:/etc/bind# named-checkzone dc.com /etc/bind/zones/db.dc.com
zone dc.com/IN: loaded serial 2
OK
root@srv-teste20:/etc/bind# named-checkzone 20.168.192.in-addr.arpa /etc/bind/zones/db.192.168.20
zone 20.168.192.in-addr.arpa/IN: loaded serial 1
OK
root@srv-teste20:/etc/bind# _
```

The terminal window has a dark background. At the bottom right, there is a taskbar with various icons and the text "Right Control".

Fonte: Elaborado pelo autor (2026).

Figura 7 – Verificação do status de execução do serviço bind9



```
root@srv-teste20:/etc/bind# systemctl status bind9
• named.service - BIND Domain Name Server
  Loaded: loaded (/lib/systemd/system/named.service; enabled; vendor preset: enabled)
  Active: active (running) since Fri 2026-10-02 00:19:33 UTC; 5s ago
    Docs: man:named(8)
  Main PID: 1862 (named)
    Tasks: 6 (limit: 2250)
  Memory: 5.4M
  CGroup: /system.slice/named.service
          └─1862 /usr/sbin/named -f -u bind

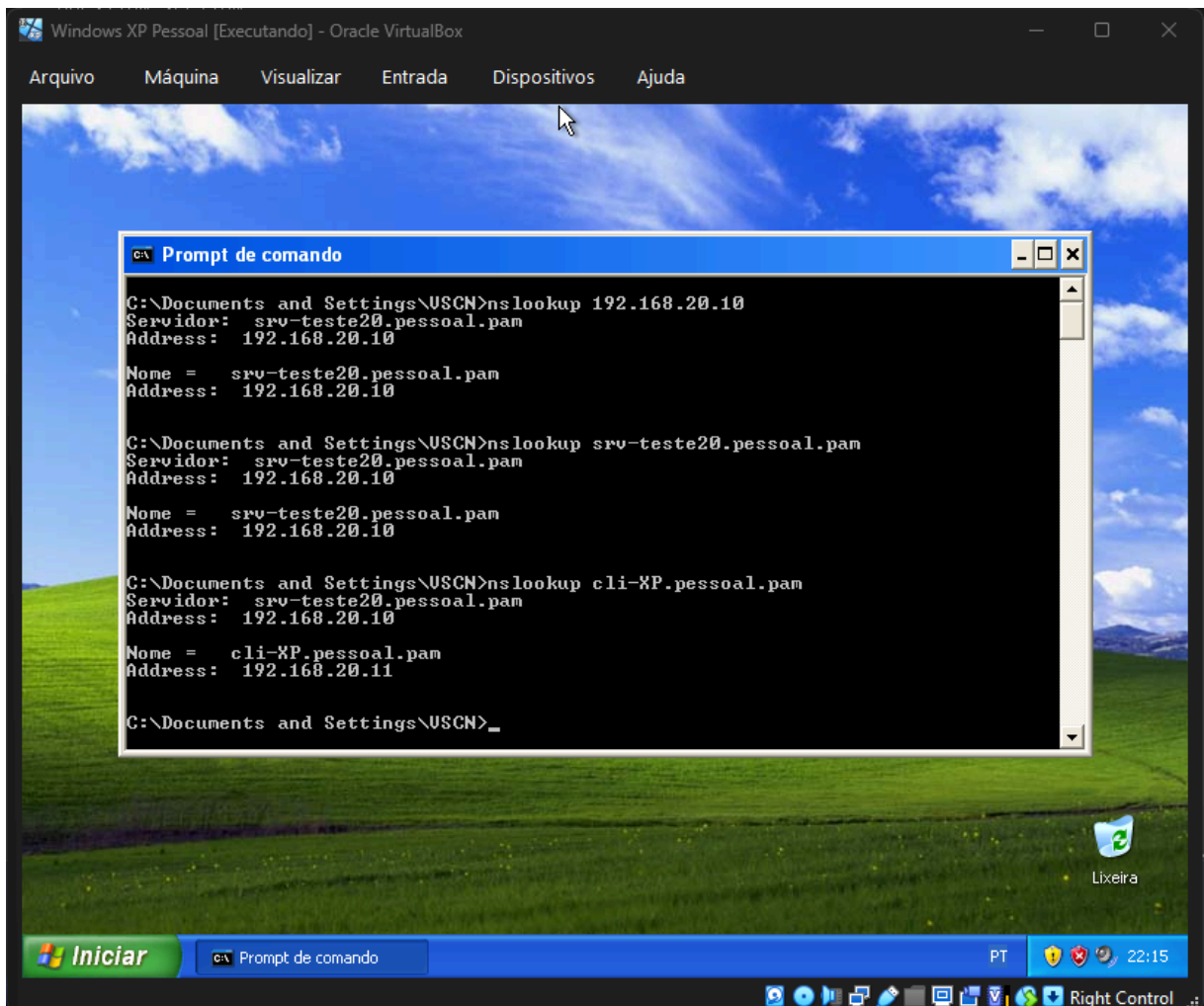
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2001:500:1::5>
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2001:500:2f::>
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2001:503:c27:>
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2801:1b8:10:>
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2001:500:2d:>
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2001:500:2::c>
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2001:7fe::53#>
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2001:dc3::35#>
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2001:500:12:>
Oct 02 00:19:35 srv-teste20 named[1862]: network unreachable resolving './DNSKEY/IN': 2001:500:9f:>

root@srv-teste20:/etc/bind# _
```

Fonte: Elaborado pelo autor (2026).

2. CLIENTE XP

Figura 8 – Testes de resolução direta e reversa com o comando *nslookup* no cliente



Fonte: Elaborado pelo autor (2026).